Paper Reading Summary

Wenfeitian Shi

February 2, 2026

Right for the Wrong Reasons: Diagnosing Syntactic Heuristics in Natural Language Inference

McCoy et al. (2019) show that many natural language inference models achieve correct predictions by relying on shallow syntactic heuristics rather than true semantic understanding. The paper identifies three common heuristics, including lexical overlap, subsequence, and constituent structure, which allow models to exploit surface-level patterns between premise and hypothesis sentences.

To analyze this behavior, the authors introduce the HANS dataset, which is designed so that heuristic-based reasoning leads to incorrect conclusions. Although many models perform well on MNLI, they fail significantly on HANS, especially on non-entailment cases. This gap reveals that standard benchmarks may overestimate a model's actual reasoning ability.

The results further indicate that adding counterexamples to the training data can reduce reliance on these heuristics. This suggests that model behavior is strongly influenced by data design, rather than reflecting an inherent understanding of language.

Shortcut Learning in Deep Neural Networks

Geirhos et al. (2020) argue that high performance in deep neural networks is often achieved through shortcut learning. In this setting, models rely on simple and task-irrelevant correlations in the data instead of learning robust and generalizable features.

Shortcut learning helps explain why models can perform well on standard benchmarks while failing under distribution shifts. The paper emphasizes that shortcuts arise from the combined effects of model architecture, training data, and optimization objectives.

As a consequence, standard evaluation benchmarks may fail to reveal such behavior. The authors also point out that interpretability methods may consistently highlight shortcut features, meaning that stable explanations do not necessarily correspond to meaningful or desired model reasoning.

The Mythos of Model Interpretability

Lipton (2018) argues that the term "interpretability" is often used ambiguously and may refer to different, sometimes conflicting, goals. The paper distinguishes between transparency, which focuses on understanding a model's internal mechanisms, and post-hoc explanations, which aim to explain model outputs after decisions are made.

Even simple models can be difficult to interpret when they involve high-dimensional features or complex preprocessing. In addition, post-hoc explanations may appear intuitive while failing to accurately reflect the true decision process of the model.

The paper concludes that claims of interpretability should clearly specify their intended purpose and evaluation criteria. Rather than treating interpretability as an inherent property of models, explanation methods themselves should be critically examined.